

CLINICAL TRIAL PROTOCOL

Code Name: GlucoControl-7 (DMT-401)
Study Title: A Phase 3, Multi-center, Randomized Trial for Type 2 Diabetes
Sponsor: Global Pharma Solutions Inc.
Revision: 4.0
Date: January 15, 2026

1. RATIONALE & OBJECTIVES

Type 2 Diabetes Mellitus affects over 530 million adults globally. Current therapies often require multiple agents and fail to achieve glycemic control in 40% of patients. GlucoControl-7 represents a novel therapeutic option with dual mechanisms of action.

Primary Objective: To demonstrate superior reduction in HbA1c compared to placebo.

Planned Target: A mean reduction of 1.5% from baseline at Week 24.

2. STUDY DESIGN

This is a double-blind, randomized, placebo-controlled study across 48 sites globally. The study will enroll 750 participants aged 18-75 with Type 2 Diabetes (baseline HbA1c 7.5-10.5%) inadequately controlled on metformin monotherapy. Study duration: 24 weeks of treatment with 4-week follow-up.

3. SAFETY MONITORING

All participants will undergo biweekly monitoring for the first month, then monthly thereafter. Special attention will be paid to renal function, liver enzymes, and signs of pancreatitis.

Adverse Event Tracking: Investigators must report all serious adverse events (SAEs) within 24 hours. The Safety Monitoring Board (SMB) will perform quarterly reviews of unblinded data.

Anticipated Effects: Based on Phase 2 data, expected incidence of mild headache is 4-6%, nausea is 3-5%, and hypoglycemic episodes (with concurrent sulfonylurea) is 2-4%. No pancreatitis cases were observed in prior phases.

4. ENDPOINTS & STATISTICS

Primary Endpoint: Change from baseline in HbA1c at Week 24.

Secondary Endpoints:

- Change in fasting plasma glucose (FPG)
- Proportion of participants achieving HbA1c <7.0%
- Change in body weight
- Change in lipid profile (LDL-C, HDL-C, triglycerides)
- Change in renal function markers (eGFR, albuminuria)